When Does a Bad Chart Actually Mislead?
Project Proposal for NLP Course, Winter 2026/27

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Abstract

Misleading charts can affect how data is understood, but not every visual flaw matters for every question. Existing research mainly focuses on detecting misleading charts or measuring how well models answer questions about them. However, it rarely examines whether a model can predict when a specific flaw will affect a particular question. This project addresses that gap by evaluating whether multimodal large language models (MLLMs) can distinguish between questions that are relevant and irrelevant to a chart flaw. We will create a benchmark using existing datasets, adding question-level relevance labels. We will compare model answers on good and misleading versions of the same charts and test different prompting methods, including flaw awareness and structured reasoning. The main deliverables will be an annotated benchmark, a comparison of model performance, and an analysis of errors across different chart flaws and question types.

1 Introduction

A bad chart can go against standard rules of good visualisation without necessarily misleading the reader about every single question they may ask about the data. For example, a truncated vertical axis may exaggerate the size of a difference between two bars, but a simpler question "Which bar is higher?" will still have the same answer. Therefore, it is important to distinguish whether a visual defect in a chart is relevant to the particular question being asked. This is especially important for multimodal large language models (MLLMs), which are increasingly expected both to interpret charts and to identify potentially misleading visualisations.

The goal of this project is to investigate whether an MLLM can determine, whether a specific visual design flaw could distort the information needed to answer a specific question. The project addresses three research questions:

1. **RQ1:** Can an MLLM distinguish questions for which a given visual defect is relevant, from questions about the same chart for which it is irrelevant?
2. **RQ2:** Do the model's question-specific relevance judgments correspond to changes in its answers between good and misleading presentations of the same data?
3. **RQ3:** How does the agreement between predicted relevance and observed answer changes vary across manipulation types and chart-question tasks?

We hypothesize that (H1) the impact of a defect is question-dependent, so the same flawed chart will contain both potentially affected and unaffected questions; and (H2) identifying a design violation, without considering the question, is insufficient to predict whether an answer will be affected. These are hypotheses to be tested, not assumed properties of every chart or model.

Chart question answering. Existing benchmarks show that chart understanding involves different skills. PlotQA uses scientific plots and real-world data to evaluate chart answering through visual information extraction and table-based reasoning (Methani et al., 2020). ChartQA extends this approach with human-written and automatically generated questions, providing a broader test of chart understanding (Masry et al., 2022). However, answering ordinary chart questions does not

necessarily mean that a model can handle misleading designs.

Bendeck et al. found that models perform differently across tasks such as identifying trends, reading exact values, and distinguishing colours, showing that accuracy depends on what a question requires (Bendeck and Stasko, 2025). Similarly, EncQA examines combinations of visual encodings and analytical tasks, finding that these combinations can matter more than model size (Mukherjee et al., 2026). These findings support evaluating the effect of visual flaws at the question level rather than treating all questions about a chart equally.

Effects of misleading visualisations. Research with human readers shows that different visual flaws can have very different effects. Rho et al. found that inverted y-axes reduced accuracy from 0.970 to 0.165, while compressed y-axes produced little change, showing that not every manipulation has the same impact (Rho et al., 2024). Pandey et al. further studied deceptive visual designs and demonstrated how changes in presentation can influence people's judgments (Pandey et al., 2015). CALVI provides a way to assess the ability to evaluate misleading graphics rather than simply read their values (Ge et al., 2023).

For MLLMs, CHARTOM compares model predictions of misleadingness with judgments from 68 human participants, finding imperfect agreement and highlighting the difficulty of estimating how misleading a chart is (Bharti et al., 2025). Together, these studies suggest that a visual flaw and its actual effect on interpretation should be studied separately.

Detecting visual flaws with LLMs. Another line of research focuses on recognising misleading charts rather than answering questions about them. Lo and Qu found that some detection prompts lead models to label too many charts as misleading, with perfect recall but precision of only 0.56–0.63 (Lo and Qu, 2025). Misviz and Misviz-synth provide large datasets for detecting visual misleaders and show that axis metadata can improve detection performance (Tonglet et al., 2026b). In related work, Tonglet et al. found that reasoning from extracted tables improved accuracy on misleading-chart questions by 15.4–19.6 percentage points, suggesting that separating data from presentation can reduce errors (Tonglet et al., 2026a). Tisha et al. evaluated both detection and explanation, reporting false-positive misleadingness judgments even for strong models (Tisha et al., 2026).

Broader studies also show why identifying design violations is not enough. Lo et al. developed a taxonomy of misleading visualisations that helps organise different sources of deception (Lo et al., 2022). Lan and Liu further examined problematic chart designs, showing the variety of ways visualisations can fail to communicate information clearly (Lan and Liu, 2025). Lisnic et al. studied misleading arguments made with charts and found that such arguments often occur even without obvious violations of design guidelines (Lisnic et al., 2023). These findings show that misleadingness cannot always be determined from visual design alone.

Benchmarks for misleading charts. Three benchmarks are especially relevant to this project. *Misleading ChartQA* (Chen et al., 2025) is a benchmark that tests whether MLLMs can answer questions despite misleading designs, but it does not systematically compare questions that are relevant and irrelevant to the same flaw. *VisDeception* allows to isolate the effects of eight visual manipulation types (Mahbub et al., 2025). The results show that some manipulations, such as inverted axes, affect many models, while others have more limited effects. However, it does not directly test whether a model can predict which questions will be affected by a given manipulation. *AttackViz* records controlled chart manipulations designed to produce specific incorrect answers and shows substantial decreases in chart-QA accuracy across several datasets (Ortiz-Barajas et al., 2026). This demonstrates that visual changes can lead models toward particular errors, but the benchmark mainly focuses on questions chosen for their vulnerability rather than comparing them with unaffected questions.

More details on these benchmarks can be found in Section 2.

Research gap. These benchmarks and previous works provide useful methods for studying misleading charts, but they do not directly evaluate whether an MLLM can predict the relevance of a flaw to individual questions before seeing its effect on the answer. This project addresses that gap by comparing the model's predicted flaw relevance with observed changes in its answers be-

tween good and misleading versions of the same data. By including both potentially affected and unaffected questions, the study aims to determine whether MLLMs can make more specific judgments than simply labelling a chart as misleading. The main contribution is an evaluation of *when a visual flaw matters for a given question*, rather than another general method for detecting flawed charts.

1.1 Concept and Work Plan

This research focuses on curating a targeted subset of deceptive charts and executing a three-stage comparative evaluation of MLLMs to investigate the mechanisms driving their reasoning during visual interpretation.

Concrete Success Deliverables

1. **Curated Context-Aware Benchmark**
   A documented subset of misleading charts extracted from source datasets, paired with validated relevant (flaw-dependent) and irrelevant (flaw-independent) questions.

2. **Comparative Evaluation**
   A report comparing model performance on relevant vs. irrelevant questions across unguided, externally structured, and CoT-prompted paradigms.

3. **Deception Error Taxonomy**
   An analysis detailing whether models fail by ignoring relevant flaws or by hallucinating interpretation changes on irrelevant questions.

The detailed project schedule, individual responsibilities, and deliverable milestones are presented in Appendix C.

1.2 Approach & Research Methodology

This research evaluates whether multimodal large language models (MLLMs) can comprehend the semantic impact of chart deception or merely detect superficial visual anomalies. As part of the experiments, basic interpretative abilities will be compared with structure-based reasoning and self-reflection, using a specially developed set of tests comprising graphs that have undergone controlled visual manipulations.

1.2.1 Experimental Pipeline

We evaluate models across three experimental configurations:

1. **Chart interpretation:** The inputs consist of the modified chart, a question about its interpretation, and a question regarding the flaw's presence and relevance. This baseline tests the central research hypothesis by measuring the model's vulnerability to unflagged visual flaws during standard interpretation tasks.

2. **Chart interpretation with flaw awarness:** The inputs consists of the modified chart, the question, and the detected flaw or chart metadata and it's relevance. This phase tests whether explicit structural grounding improves relevance judgments and enables the model to correct its reasoning.

3. **Chain-of-Thought (CoT) chart interpretation:** The inputs are the modified chart and question alongside a staged reasoning prompt (e.g., "First, identify visual encoding distortions and their relevance for chart interpretation. Second, answer the question."). This tests whether the model can act as its own autonomous supervisor to prevent misleading information from being introduced.

The overview of the experimental pipeline is presented in Figure 1.

1.2.2 Evaluation Metrics

To assess model performance across the experimental setups, we will calculate standard classification metrics alongside two other metrics designed to investigate the semantic impact of chart deception.

Basic Classification Metrics: Standard classification metrics, including accuracy, precision, recall, and F1-score, will be used to evaluate the models' ability to classify visual flaws as relevant or irrelevant for a given chart question. These measures will provide an overall assessment of the reliability and effectiveness of the models' relevance judgments.

General Accuracy (GA): The fundamental performance of the models will be evaluated using standard accuracy, defined as the overall percentage of questions answered correctly.

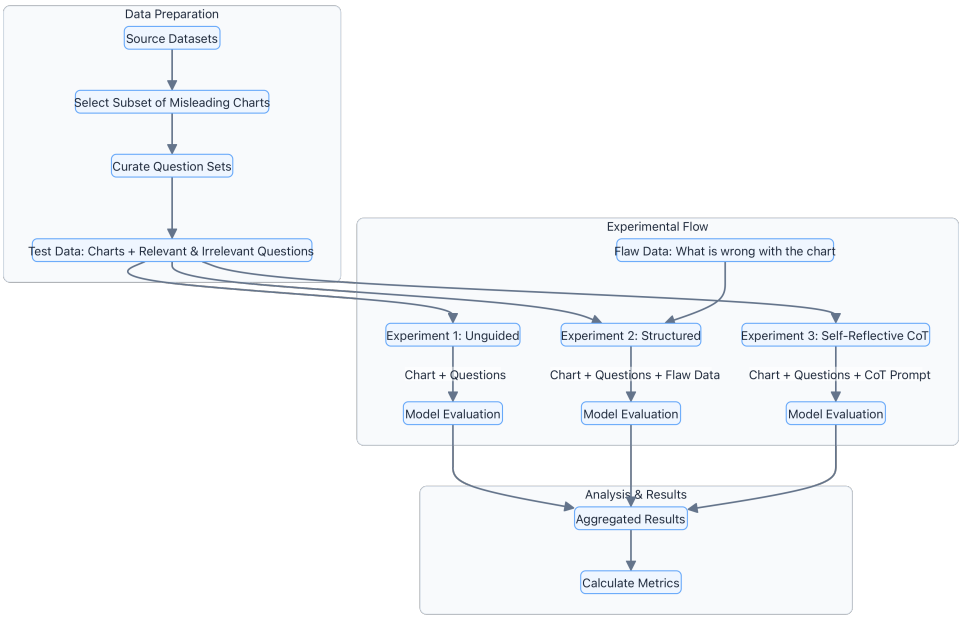


Figure 1: Experimental pipeline schema

Regression Rate (Over-correction): This metric tracks the percentage of irrelevant questions that the model answered correctly in the unguided baseline (Experiment 1), but flipped to incorrect in Experiments 2 or 3. This assesses whether exposing the model to flaw metadata causes it to second-guess and ruin an accurate answer.

Deception Impact (DI): This metric isolates the model's vulnerability to visual deception. It is calculated by subtracting the model's accuracy on relevant questions (where the answer relies on the deceptive feature) from its accuracy on irrelevant questions (where the flaw does not matter).

Recovery Rate (RR): This metric verifies the practical utility of providing structural metadata (Experiment 2) or prompting self-reflection (Experiment 3). It measures the percentage of relevant questions that scored as incorrect in Experiment 1 but were answered correctly in subsequent experiments. This determines whether flaw awareness corrects model's reasoning or serves as unused context.

2 Datasets and Their Status

The project will use three existing chart-question answering datasets: *Misleading ChartQA* (Chen et al., 2025), *AttackViz* (Ortiz-Barajas et al., 2026), and *VisDeception* (Mahbub et al., 2025). Our proposed contribution is a new question-level annotation layer that specifies whether a particular visual defect is relevant to answering a given question.

Table 1 summarizes the technical details of selected datasets. None of the three datasets directly provides the balanced, question-specific defect-relevance labels required by this project. AttackViz and VisDeception already offer controlled chart pairs; good counterparts for selected Misleading ChartQA items may need to be reconstructed from their chart code and underlying data. The additional question-level relevance annotations are therefore part of the processing pipeline, not part of the **READY** datasets.

3 Data Statement

The project uses publicly available chart datasets: Misleading ChartQA, AttackViz, and VisDeception. AttackViz and VisDeception are distributed under GPLv3; Misleading ChartQA redistribution requires citation of original paper. We will create question-level relevance annotations without collecting participant or other personal data. Charts will be screened for identifying information, which will be excluded or redacted where necessary. We will release only materials permitted by the respective licences.

Table 1: Existing chart datasets - sizes, labels, licences, and acquisition status. Dataset-level access is distinct from the new relevance annotations, which remain to be collected.

Dataset	Size and class balance	Labels and annotations	Licence and availability
Misleading ChartQA (Chen et al., 2025)	3,026 curated examples in the paper; approximately 3,060 in the public repository. The original benchmark contains misleading charts, rather than a balanced good/misleading classification.	21 misleader types; 10 chart types; multiple-choice questions, correct answers, misleading distractors, explanations, underlying CSV data, and chart code.	**READY** for research access (public GitHub repository). Requires paper citation.
AttackViz (Ortiz-Barajas et al., 2026)	18,247 instances in the public release: 6,059 train, 6,146 validation, and 6,042 test. Contains good and misleading examples, relatively balanced.	Binary chart label (0 = faithful, 1 = misleading); 11 design-manipulation categories; questions, good and manipulated answers, and chart specifications.	**READY** (public Hugging Face dataset). Combined dataset: GPLv3; framework code: Apache 2.0.
VisDeception (Mahbub et al., 2025)	1,600 chart images forming 800 pairs: 800 good and 800 misleading (50/50 balance); 100 pairs per manipulation type.	Good/misleading pair identity, one of eight manipulation categories, and evaluation questions about perceived magnitude, direction, or trends.	**READY** (public GitHub repository); GPL-3.0.

4 Contribution Table and LLM-use

Author	Completed work (Phase 1)	Hours
Zofia Kamińska	literature review (5 works), report structuring, repository setup, literature review documentation	5
Karolina Dunal	literature review (5 works), dataset descriptions, experimental setup documentation	5
Mateusz Deptuch	literature review (5 works), dataset descriptions, experimental setup conceptualisation	5

LLM use declaration. ChatGPT (OpenAI; GPT-6) was used to assist in literature discovery and draft the research proposal. As the authors, we remain responsible for verifying original papers, results, references, dataset rights, and all submitted work.

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A List of abbreviations

Abbreviation	Meaning
AI	Artificial intelligence
CQA	Chart question answering
EM	Exact match (evaluation metric)
F1	F1 score (harmonic mean of precision and recall)
FACT	Factual chart-reading component of CHARTOM
GPT	Generative pre-trained transformer
LLM	Large language model
ML	Machine learning
MLLM	Multimodal large language model
MRFT	Model Reasoning Failure Taxonomy
MCQ	Multiple-choice question
MIND	Human-perception prediction component of CHARTOM
N/R	Not reported
NLP	Natural language processing
PDF	Portable Document Format
pp	Percentage points (when comparing accuracy or performance)
QA	Question answering
SBERT	Sentence-BERT (sentence-level Bidirectional Encoder Representations from Transformers)
SOTA	State of the art
VLAT	Visualization Literacy Assessment Test
VLM	Vision-language model
Abbreviations and short names used as benchmark, method or model identifiers	
CHARTOM	Visual theory-of-mind benchmark for misleading charts
ChartQA	Chart question answering benchmark
EncQA	Benchmark for visual encodings in chart question answering
LLaVA	Large Language and Vision Assistant (model family)
SAN	Stacked Attention Network (model baseline)
T5	Text-to-Text Transfer Transformer (model family)
VL-T5	Vision-language variant of T5
VOES	Model baseline named VOES in the PlotQA comparison

B Grouped literature review summary

B.1 Misleading Charts and AI

Table 3: Sources describing directly the concept of misleading charts and AI. Unspecified dataset name means authors used a custom benchmark

Source	Methodologies and Contibutions	Dataset (evaluation procedure)	Method (models / people)	Key results
1. (Chen et al., 2025)	Creates misleading chart MCQs with correct and visually tempting wrong answers. Evaluates a region-aware reasoning pipeline that first identifies suspicious chart regions.	Misleading ChartQA: 3,026 examples, 21 flaw types, 10 chart formats; full benchmark evaluation, no conventional train/test split.	24 MLLMs: GPT, Claude, Gemini, Qwen, InternVL, DeepSeek.	GPT-4o baseline 34.93%; best region-aware pipeline 51.94% with InternVL3-78B-MPO, evaluated on the published benchmark.
2. (Mahbub et al., 2025)	Pairs real and misleading charts while preserving the underlying data. Evaluates whether visual manipulations affect VLM responses without explicitly warning the models.	VisDeception: approximately 800 normal and 800 misleading charts across eight manipulation types; paired evaluation using the Wilcoxon signed-rank test.	10 VLMs (5 closed-source, 5 open-source).	Significant effects ($p < 0.05$) were observed for inverted axes (10/10 models), aspect-ratio distortions (9/10), truncated axes (8/10), and inappropriate continuous encodings (8/10). Distorted projections and inappropriate categorical encodings affected only 3/10 models each, indicating that susceptibility varies substantially across manipulation types.
3. (Tonglet et al., 2026a)	Tests six inference-time mitigation strategies, including warnings, metadata extraction, table-only reasoning and chart redrawing. Shows that removing dependence on the misleading visual encoding improves QA accuracy.	143 misleading + 124 faithful evaluation charts; separate ChartQA test = 2,500 questions.	19 MLLMs, with additional text-only components for some strategies.	Table-based QA improves misleading-chart accuracy by 15.4–19.6 pp, depending on the experimental comparison.
4. (Bharti et al., 2025)	Separates factual chart reading (FACT) from predicting how humans will interpret a chart (MIND). Introduces a human-calibrated misleadingness score rather than relying only on model QA accuracy.	112 charts, comprising 56 faithful/manipulated pairs; 14 manipulations; 68 human participants for calibration; evaluation-only benchmark.	10 MLLMs across several model families; human respondents.	All 10 MLLMs struggled to predict human misleadingness (HMI), with MAE ranging from 0.24 (Gemini 2.5, $\sigma = 0.16$) to 0.46 (Llama 3, $\sigma = 0.27$). GPT-4o achieved MAE 0.35. Even the best model showed substantial disagreement with human judgments.
5. (Lo and Qu, 2025)	Compares nine prompting strategies for detecting progressively more flaw types. Finds that prompting can encourage unnecessary issue reporting, producing misleading false positives.	Experiments of 54, 84 and 150 charts; final split 75 development / 75 test, with 63 misleading and 12 valid in the test.	GPT-4 Vision, Copilot, Gemini Pro Vision, LLaVA-NeXT-7B.	Models tended to over-report chart flaws. In Experiment 1, Prompt #1 achieved perfect recall (1.00) across all models, but relatively low precision (0.56–0.63), indicating frequent false positives. Performance varied with prompting; on the final test set ($n = 75$), Prompt #7 achieved F1 scores of 0.79 (GPT-4), 0.58 (Copilot), 0.67 (Gemini), and 0.03 (LLaVA).
6. (Tonglet et al., 2026b)	Introduces real-world and synthetic flaw-detection benchmarks. Compares zero-shot VLMs, a rule-based linter and classifiers supplied with extracted axis metadata.	Misviz 2,604 charts: dev 5%, val 15%, test 80%. Misviz-synth 57,665: train-large 80.1%, train-small 9.6%, dev 3.1%, val 3.2%, test 4.1%.	GPT-4.1, o3, Gemini 2.5 Flash Lite, Qwen2.5-VL, InternVL3, DePlot and classifiers.	Misviz test: GPT-o3 multi-label F1 80.0%, EM 58.8%. Synthetic test: image-plus-axis classifier F1 65.2%, EM 69.5%.
7. (Tisha et al., 2026)	Introduces a three-stage workflow for chart extraction, misleadingness detection, and explanation, alongside a taxonomy of 10 model reasoning failure types (MRFT).	500 real-world charts: 400 collected from social media (Reddit, Twitter) and 100 from student contributions; manually annotated using Lo et al.'s taxonomy. Evaluation split not reported.	GPT-5.5 and Gemini 3.1 Pro.	False-positive misleading detection was the most common failure (13.9% GPT-5.5; 15.3% Gemini). Human agreement (α) was higher for binary detection (0.74/0.59) than issue-tag classification (0.33/0.51), highlighting difficulty identifying specific flaws. GPT-5.5 showed stronger reasoning-error agreement (0.85 vs. 0.76).

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Table 3 (continued)

Source	Methodologies and Contibutions	Dataset (evaluation procedure)	Method (models / people)	Key results
8. (Ortiz-Barajas et al., 2026)	Automatically generates chart manipulations designed to induce specific wrong answers. Introduces AttackViz and tests targeted effects on models and humans, including fine-tuning-based defenses.	AttackViz (own): approximately 18,200 instances; current public release: 18,247, train 6,059 / val 6,146 / test 6,042. Published domain evaluations use PlotQA, ChartQA and ChartX.	16 VLMs including GPT-4o, Gemini, Claude, InternVL, Ovis and LLaVA; SBERT retrieval and Qwen fine-tuning.	ChartAttack reduced VLM QA accuracy by 17.2 pp on PlotQA, 11.5 pp on ChartQA, and 12.3 pp on ChartX. In-domain, 11.2% of initially correct answers shifted to attacker-intended incorrect answers. Effects varied by manipulation: inappropriate stacking caused a 41.5 pp accuracy drop, while ineffective color schemes had negligible impact. Human QA accuracy fell from 88.3% to 71.9% ($n = 48$, $p < 0.001$). Fine-tuning improved in-domain accuracy by 10.51 pp but reduced performance on clean charts.

B.2 Human Based Evaluations

Table 4: Human perception and visualization methodology

Source	Methodologies and Contibutions	Dataset (evaluation procedure)	Method (models / people)	Key results
9. (Ge et al., 2023)	Develops a psychometrically validated assessment of critical visualization literacy. Connects chart design, question type and wrong-answer mechanisms using item response theory.	497 human participants in test development; final 45-item bank, with a shorter assessment also proposed. No ML split.	Human	Final item-bank reliability ω = 0.81 across 45 items; no VLM accuracy metric.
10. (Rho et al., 2024)	Compares identical questions across controlled faithful/misleading chart pairs. Demonstrates that some manipulations severely affect human answers while others have limited effects.	112 charts, 56 pairs, 14 manipulation types; human experimental conditions, no ML split.	Human	Human accuracy, faithful $\rightarrow$ misleading: inverted y-axis 0.970 $\rightarrow$ 0.165; irregular x-axis 0.952 $\rightarrow$ 0.080; compressed y-axis 0.904 $\rightarrow$ 0.926.
11. (Pandey et al., 2015)	Runs randomized human experiments on visual distortions affecting perceived magnitude or trend direction. Separates questions about how much from questions about what direction.	330 recruited participants, two studies; quality-filtered inversion experiment 40 control / 38 deceptive.	Human	Inversion experiment: 97.5% correct in the control group versus 18.42% in the deceptive group
12. (Lisnic et al., 2023)	Analyzes chart-related arguments in authentic social-media posts. Shows that unsupported conclusions often arise without recognizable chart-design violations.	Approximately 10,000 COVID-19 Twitter posts containing charts; contextual human annotations; no ML split.	Human	Among posts offering chart interpretations, about 84% involved misleading reasoning; only 11% of those misleading posts violated common chart-design guidelines.
13. (Lo et al., 2022)	Builds a taxonomy through open coding of real-world misleading graphics. Identifies design and reasoning problems beyond the usual list of axis-scale violations.	129,125 images collected; 6,500 examined; 1,143 issue-tagged examples; no ML train/test.	Human	74 issue types organized into 12 categories, established through qualitative analysis rather than predictive evaluation.
14. (Lan and Liu, 2025)	Develops a task-associated flaw taxonomy from public critiques. Investigates why design problems occur using focus groups, creating a more explanatory framework than a simple checklist.	2,227 flawed charts; five focus groups; no ML split.	Human	76 specific flaws, 10 subcategories, three broad categories, seven identified causes and nine research opportunities.

B.3 General Chart Understanding and QA Benchmarks

Table 5: General chart understanding and QA benchmarks

Source	Methodologies and Contibutions	Dataset (evaluation procedure)	Method (models / people)	Key results
15. (Bendeck and Stasko, 2025)	Evaluates visualization literacy using established questions and repeated model calls. Shows that recognizing broad patterns is easier for GPT-4 Vision than reliable value retrieval and color discrimination.	VLAT 53 questions; ChartQA evaluation subset 629 questions; no training.	GPT-4 Vision, three runs per question.	On VLAT n=53, guessing-adjusted score 19.67; raw correct responses by difficulty: easy 12/17, moderate 10/19, hard 6/17.
16. (Mukherjee et al., 2026)	Systematically varies analytical task and chart encoding. Demonstrates that model performance depends on both dimensions and does not consistently improve with parameter count.	2,076 QA pairs; six encodings $\times$ eight task types; evaluation benchmark with no verified standard training split.	Nine VLMs including GPT-4o, o1, InternVL2, Molmo, Phi-3.5-Vision and ChartGemma.	Major task-by-encoding variation and non-monotonic size effects; exact per-model EncQA accuracy values require transcription from the paper's result figures.
17. (Masry et al., 2022)	Combines human-written and automatically generated chart questions. Introduces models that integrate extracted chart tables with visual features for logical reasoning.	32,719 QAs; train 28,299, validation 1,920, test 2,500 (1,250 human and 1,250 generated).	VisionTaPas, VL-T5, T5 and ChartOCR components.	On ChartQA test n=2,500, PlotQA-pretrained VL-T5 51.84% overall, 40.08% human subset, 63.60% generated subset.
18. (Methani et al., 2020)	Generates large-scale analytical questions over scientific plots using authentic data. Proposes a hybrid system combining visual extraction with structured-table QA for open-vocabulary numeric answers.	224,377 plots and about 28.9 million QAs; benchmark train/validation/test partitions.	Hybrid chart/table QA, VOES, SAN and other baselines.	On PlotQA test, hybrid model 22.52% accuracy versus 18.46% VOES and 7.76% SAN.

C Work Plan and Responsibilities

Table 6 presents the project schedule and the planned division of responsibilities among the three team members.

Table 6: Project timeline and individual responsibilities.

Period / Milestone	Zofia Kamińska	Karolina Dunal	Mateusz Deptuch
4 Nov 2026 Phase 1	Finalize SOTA review, research questions, and proposal structure.	Finalize experimental methodology, pipeline, and evaluation metrics.	Conceptualise experimental setup, review literature.
5–11 Nov Dataset preparation	Define subset selection criteria and question-relevance annotation guidelines.	Download and inspect source datasets; select chart pairs and classify flaws.	Prepare data-loading scripts, experiment environment, and repository.
12–17 Nov Proof of concept	Validate relevant and irrelevant questions and prepare presentation content.	Create an initial annotated subset and verify chart-question pairs.	Implement baseline inference and preliminary relevance-prediction tests.
18 Nov Mid-term delivery	Present research motivation, SOTA, and proposed methodology.	Present dataset examples and annotation procedure.	Demonstrate the proof of concept and preliminary results.
19–25 Nov Experiments 1–2	Review relevance labels and define comparison between predicted relevance and observed answer changes.	Finalize the curated subset and prepare structured flaw metadata.	Run unguided and flaw-aware experiments on good and misleading chart pairs.
26 Nov–2 Dec Phase 2	Prepare the Phase 2 report, interpret initial findings, and coordinate rebuttals.	Document dataset construction, annotation quality, and experimental examples.	Run CoT experiments, calculate initial metrics, and finalize source code.
3–20 Dec Evaluation and analysis	Analyze agreement between predicted relevance and observed answer changes.	Analyze false positives, false negatives, and differences across manipulation types.	Verify experimental results, compare model configurations, and generate plots.
21 Dec–12 Jan Final preparation	Integrate findings into the final report and prepare the final presentation.	Finalize error taxonomy, result tables, and visual examples for the report.	Clean and document code, reproduce results, and prepare the final video.
13 Jan 2027 Phase 3	Joint submission of final report, presentation, video, and project artifacts.		

Expected deliverables. Phase 1 delivers the research proposal and SOTA analysis. The mid-term milestone demonstrates an initial annotated dataset and a working proof of concept. Phase 2 delivers the report, source code, rebuttals, and initial results from all three experimental configurations. Phase 3 delivers the final evaluation, error analysis, reproducible code, report, presentation, and video.